WhatNext Vision Motors: Shaping the Future of Mobility with Innovation and Excellence

1. Introduction

The automotive industry is rapidly evolving with the integration of advanced technologies such as Artificial Intelligence, Electric Vehicles (EVs), and smart mobility solutions. WhatNext Vision Motors is designed as an innovative platform to transform traditional transportation into a smart, efficient, and sustainable system.

This project focuses on developing a modern mobility solution using digital technologies to improve vehicle management, user experience, and environmental sustainability. The system leverages automation, real-time data, and intelligent decision-making to redefine the future of transportation.

2. Objective

To develop a smart mobility management system

To manage vehicles, users, and services efficiently

To promote sustainable transportation solutions

To ensure data accuracy using validation mechanisms

To provide insights using reports and dashboards

3. User Story

In today's fast-paced world, users face challenges in managing vehicles, tracking maintenance, and accessing smart mobility services. WhatNext Vision Motors provides a unified digital platform where users can monitor vehicle performance, manage services, and make informed decisions.

The system enhances convenience, reduces manual effort, and promotes eco-friendly transportation through intelligent solutions

4. System Modules

4.1 User Module

Stores user details such as name, contact information, and preferences.

4.2 Vehicle Module

Maintains vehicle information including type (EV/Fuel), model, and status.

4.3 Service Module

Tracks vehicle servicing details such as service date, type, and cost.

4.4 Charging/Fuel Module

Monitors charging or fuel usage, including energy consumption and cost.

4.5 Smart Analytics Module

Stores and analyzes data such as vehicle performance and usage patterns.

5. System Architecture

The system is implemented using modern CRM/Cloud technologies and includes:

Custom Objects

Relationships between entities

Validation Rules

Flow Automation

Optional Apex Logic

Reports and Dashboards

6. Project Development Steps

1. Designed system architecture
2. Created custom objects (User, Vehicle, Service, Charging/Fuel, Analytics)
3. Established relationships between modules
4. Built user interface and navigation
5. Designed layouts for better usability
6. Implemented validation rules
7. Developed automation flows

8. Generated reports and dashboards

9. Conducted testing and validation

7. Validation Rule

A validation rule is implemented to ensure valid service cost data.

Formula:

Service_Cost__c < 0

Error Message: Service cost cannot be negative.

8. Flow Automation

A record-triggered flow is created to:

Check duplicate service entries

Validate vehicle and service date

Prevent incorrect data entry

Display error messages when needed

9. Reports and Dashboard

Reports Created:

Vehicle Usage Report

Service History Report

Dashboard Components:

Vehicle Type Distribution (Pie Chart)

Service Frequency (Bar Chart)

Energy Consumption (Metric)

10. Testing

Test Cases:

1. Add vehicle → Successful

2. Invalid service cost → Error displayed

3. Duplicate service entry → Error shown

4. Dashboard → Displays correct insights

11. Advantages

Enhances mobility efficiency

Reduces manual tracking effort

Ensures accurate data management

Supports smart decision-making

Encourages sustainable transportation

12. Limitations

Requires internet connectivity

Dependent on platform infrastructure

Limited features without advanced integration

13. Future Enhancements

Integration with IoT-enabled vehicles

Mobile application development

AI-based predictive maintenance

Smart route optimization

14. Team Structure and Work Distribution

Team Structure

Kailashini

Kameshwaran

Vijayalakshmi

Vasan

Work Distribution

Member 1 – Development

Created system modules

Implemented validation and automation

Member 2 – UI/UX Design

Designed interface and navigation

Improved user experience

Member 3 – Reports & Dashboard

Generated reports

Designed dashboard visuals

Member 4 – Testing & Deployment

Performed system testing

Managed deployment

15. Conclusion

WhatNext Vision Motors successfully demonstrates how modern technology can transform the mobility sector. The system improves efficiency, enhances user experience, and promotes sustainable transportation through smart and innovative solutions.

16. References

Cloud/CRM Documentation

Smart Mobility Concepts

Online Learning Resources

Author

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